Neural Networks

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What we will cover?

Neurons or Perceptrons

Activation Functions

Cost Function

Backpropogation

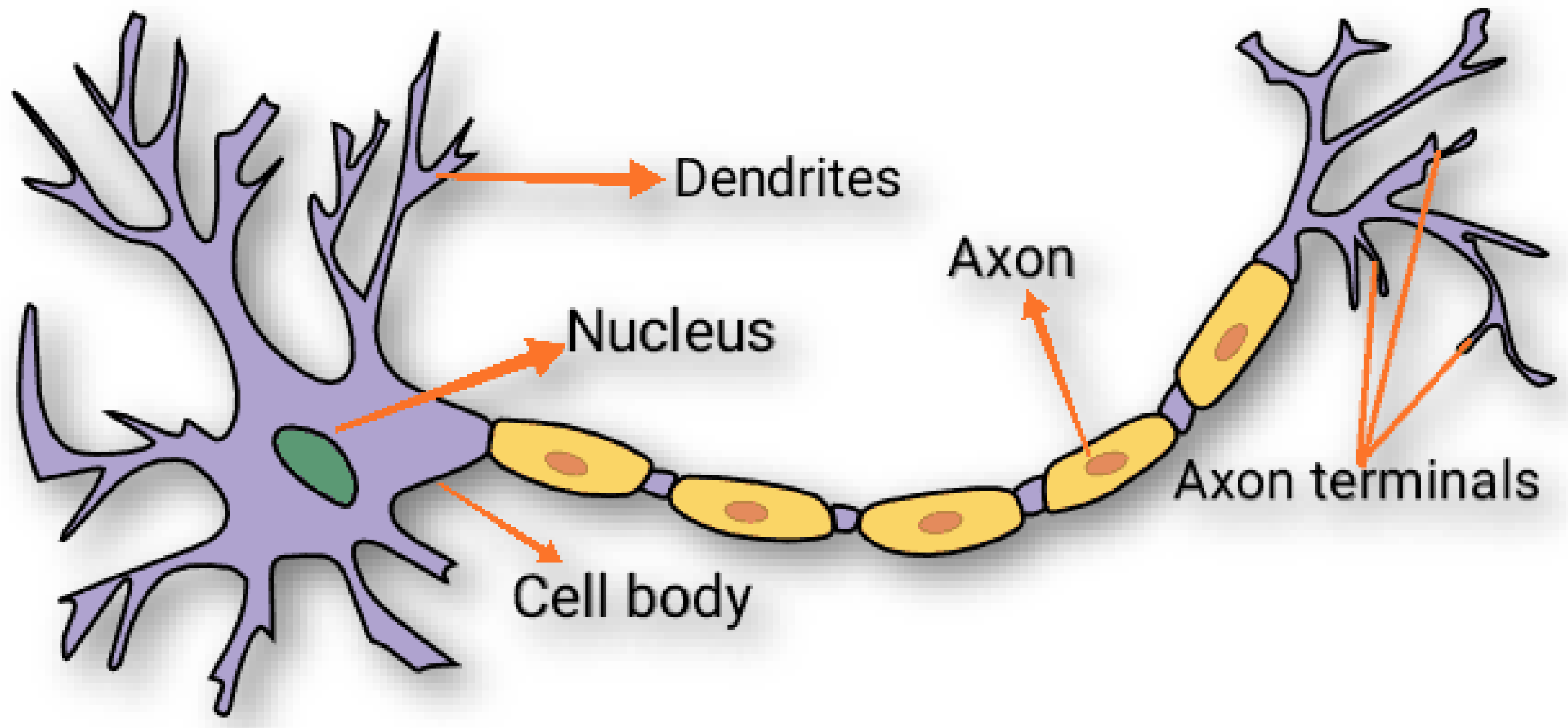
Gradient Descent

Neurons / Perceptron

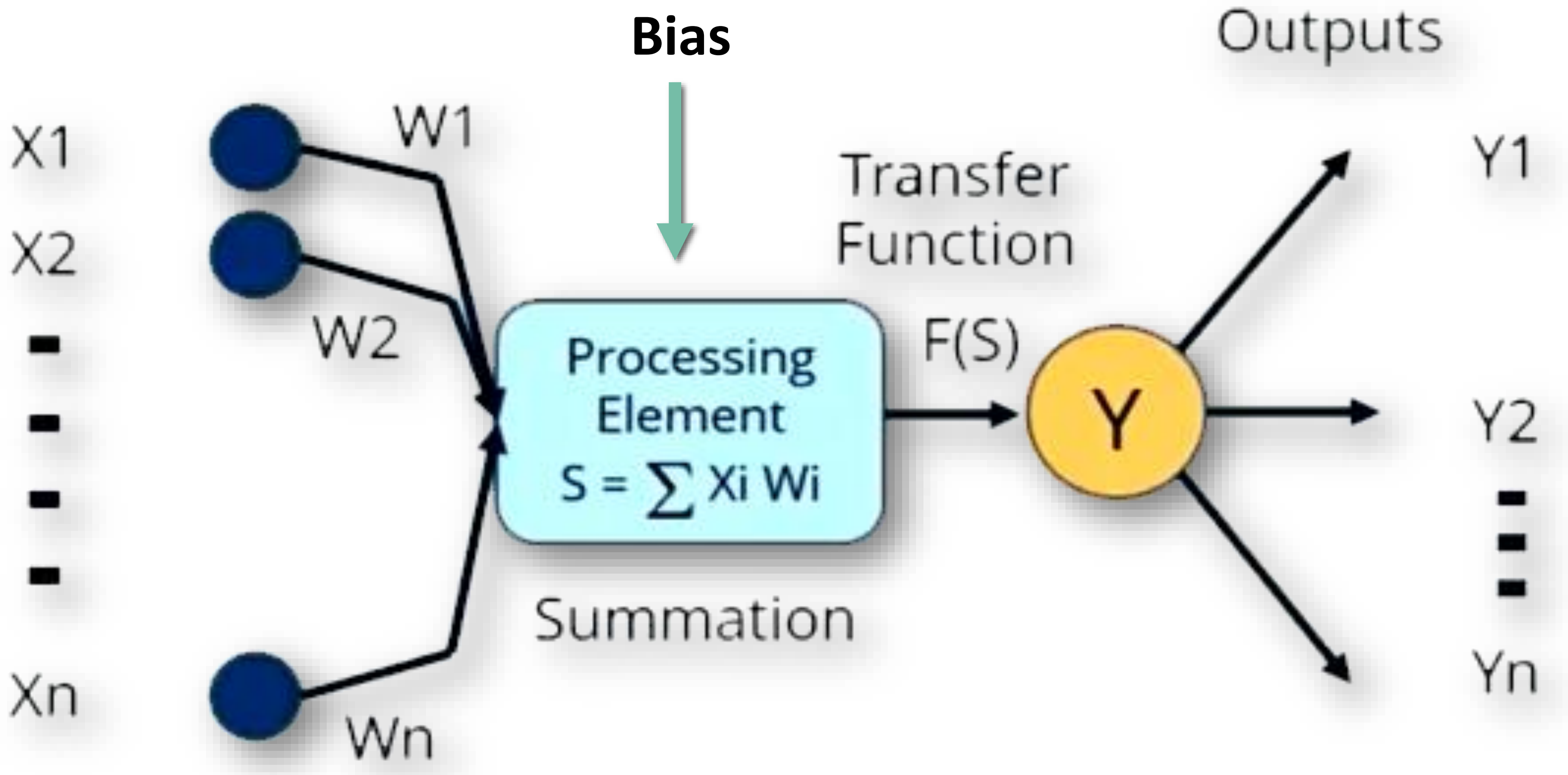
- Very important part of brain through which the signals are transmitted.
- **Artificial Neural Network** is based on Neural Biological Systems.
- ANN is a software based approach to replicate the biological neurons.

Let’s discuss about Biological Neurons !!

Biological Neurons



Artificial Neurons



Perceptron Mathematical Model

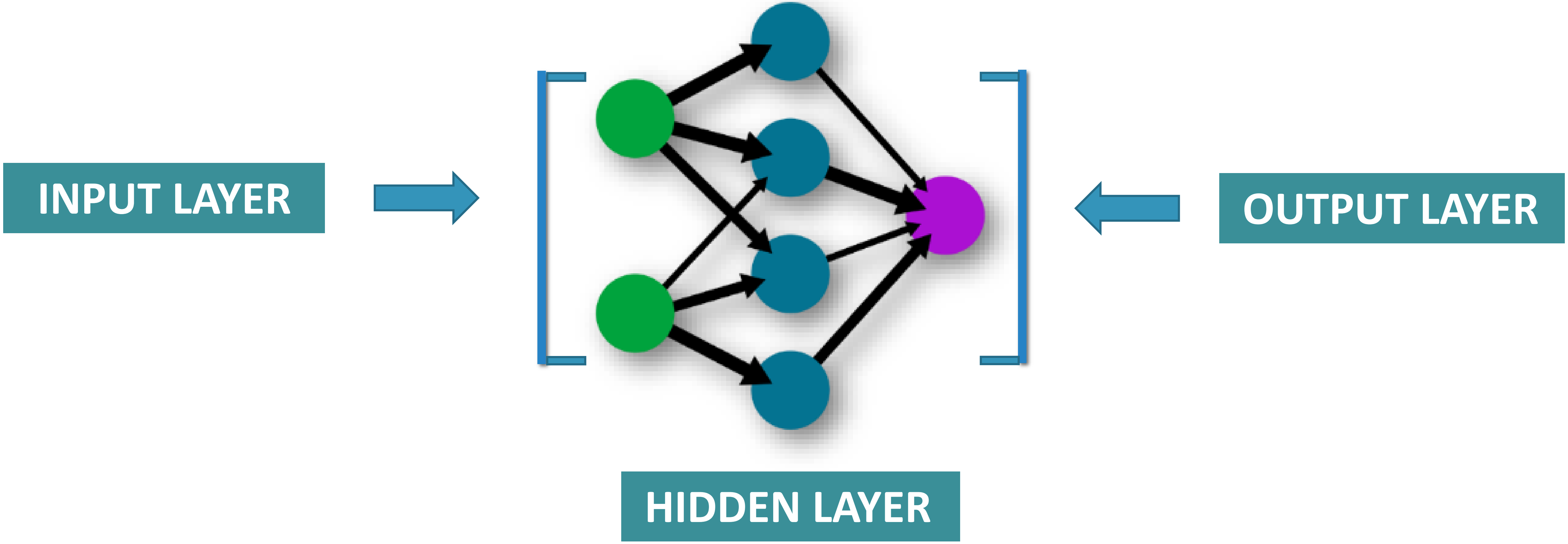
$$f(x) = b_0 + \sum_{i=0}^{n} (b_i * W_i)$$

b_i = inputs

W_i = Weights

b_0= bias

Multiple Perceptron Networks

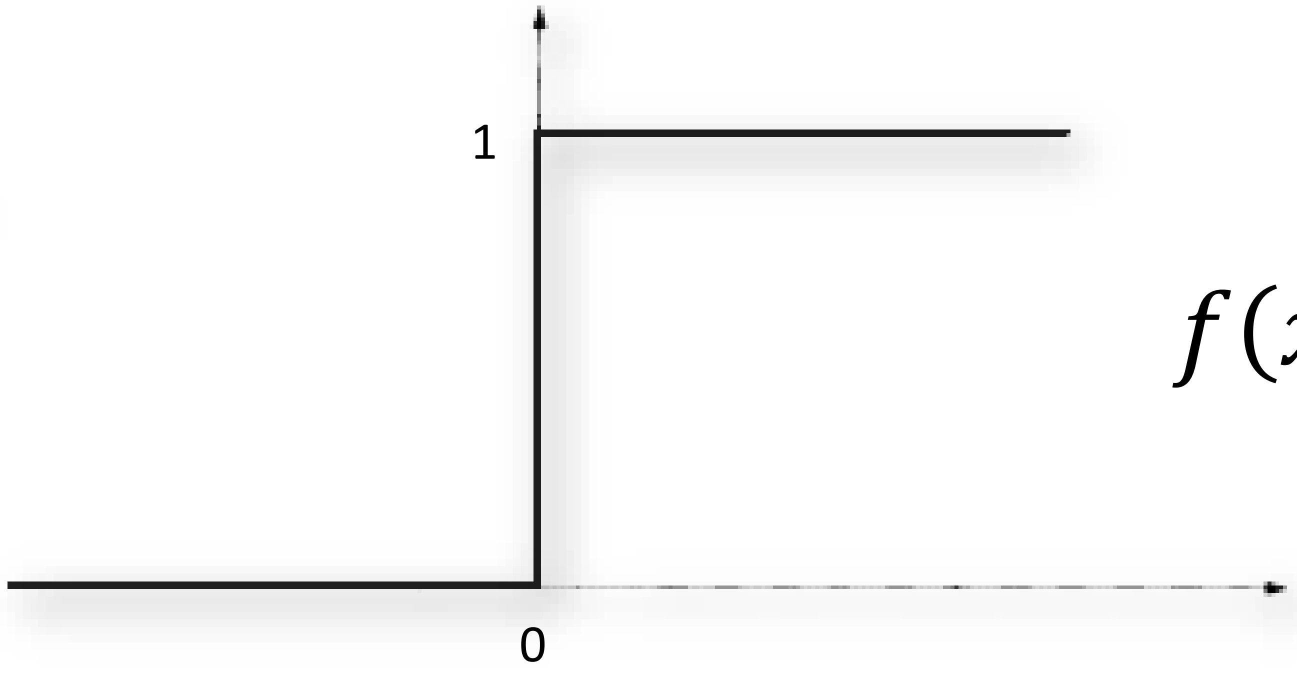


Activation Functions

- Main role is to calculate and decides the output.
- Types –
  - Step Function
  - Sigmoid Function
  - Hyperbolic Tangent
  - Rectified Linear Unit (ReLU)

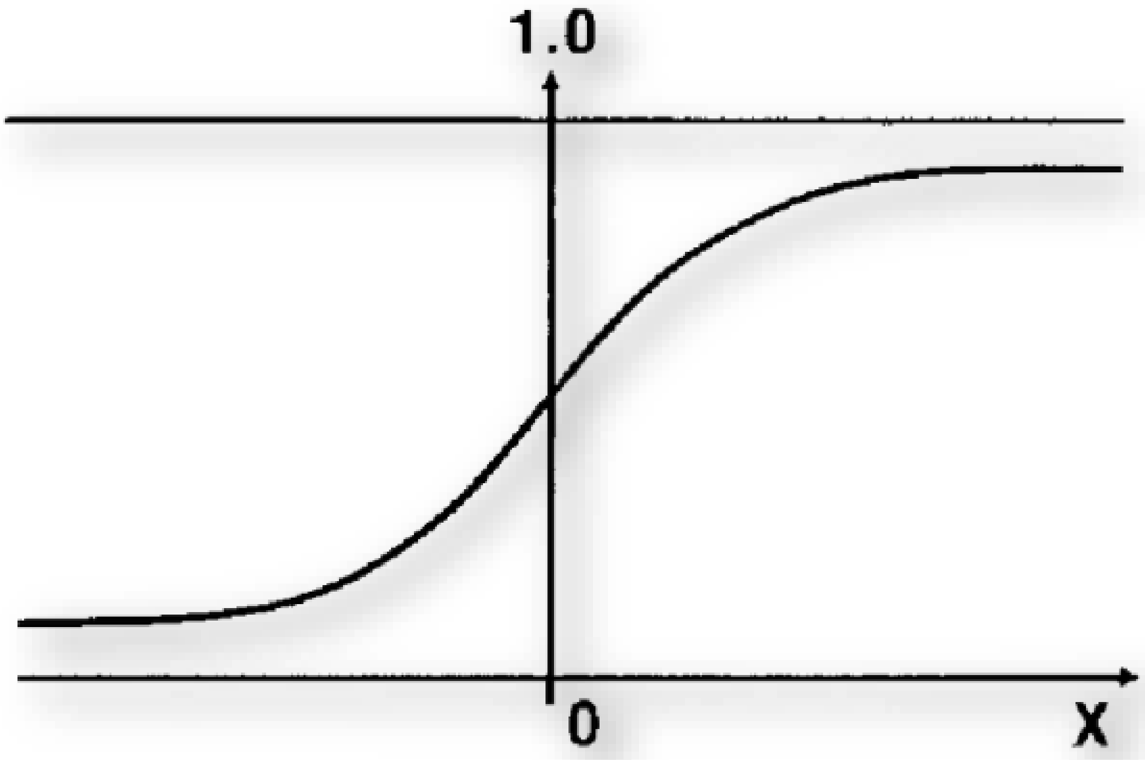
Let’s discuss about these one by one !!

Step Function



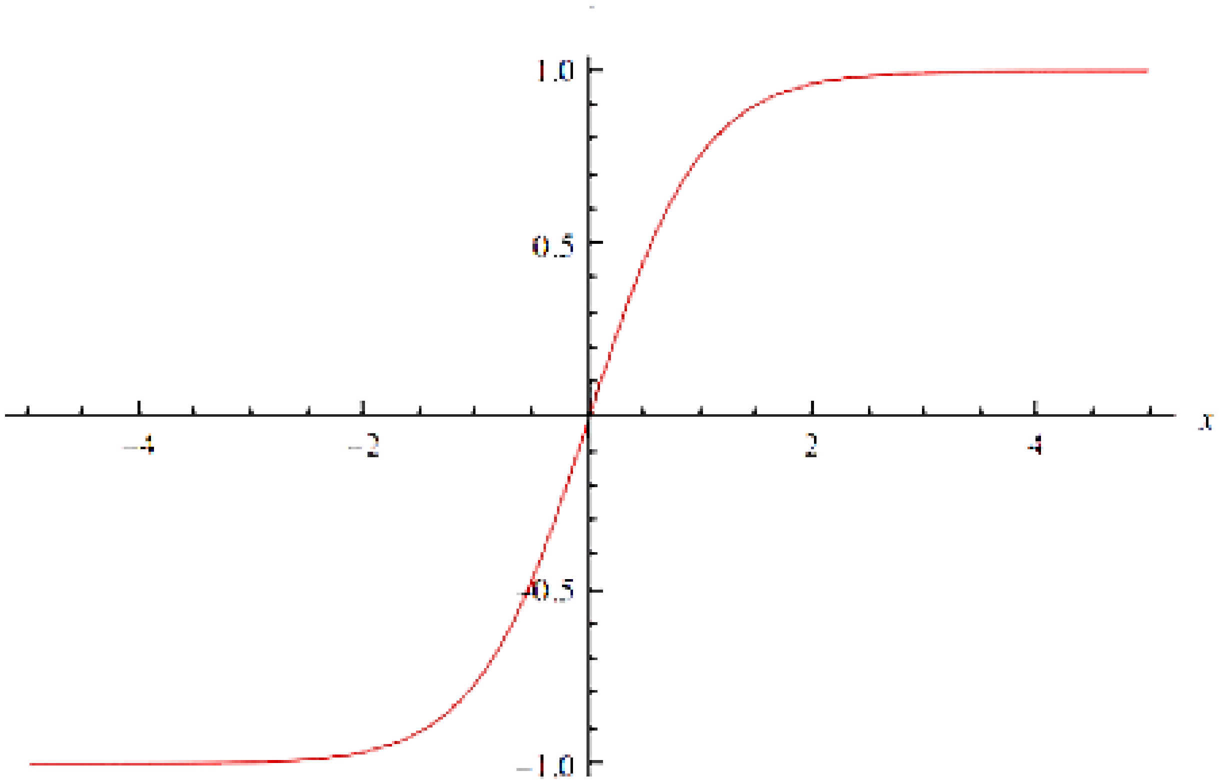
$$f(x) = \begin{cases} 0 \ if \ x < 0 \\ 1 \ if \ x \geq 0 \end{cases}$$

Sigmoid Function



$$f(x) = \frac{1}{1 + e^{-(x)}}$$

Hyperbolic Tangent Function

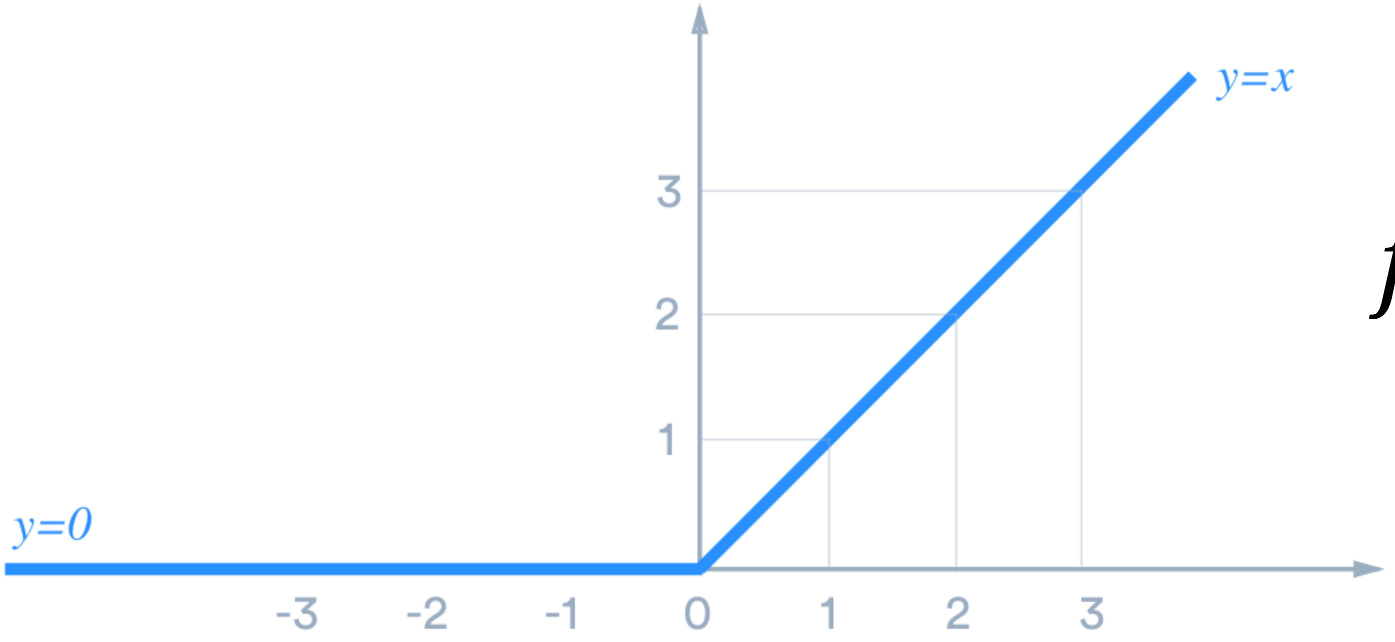


$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

$$\sinh x = \frac{e^{x} - e^{-x}}{2}$$

$$\tanh x = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

Rectified Linear Unit (ReLU) Function



$$f(x) = \max(0, x)$$

Cost Function

- It calculates difference between the actual and predicted values, i.e. it calculates the error.

$$C = \frac{\sum (y - a)^2}{n}$$

y = actual values

a = predicted values

- It has one drawback as it slows down the learning of neural network, So in this case we use "Cross Entropy"

Cross Entropy

- It reduces the cost function and increases the learning of neural network.

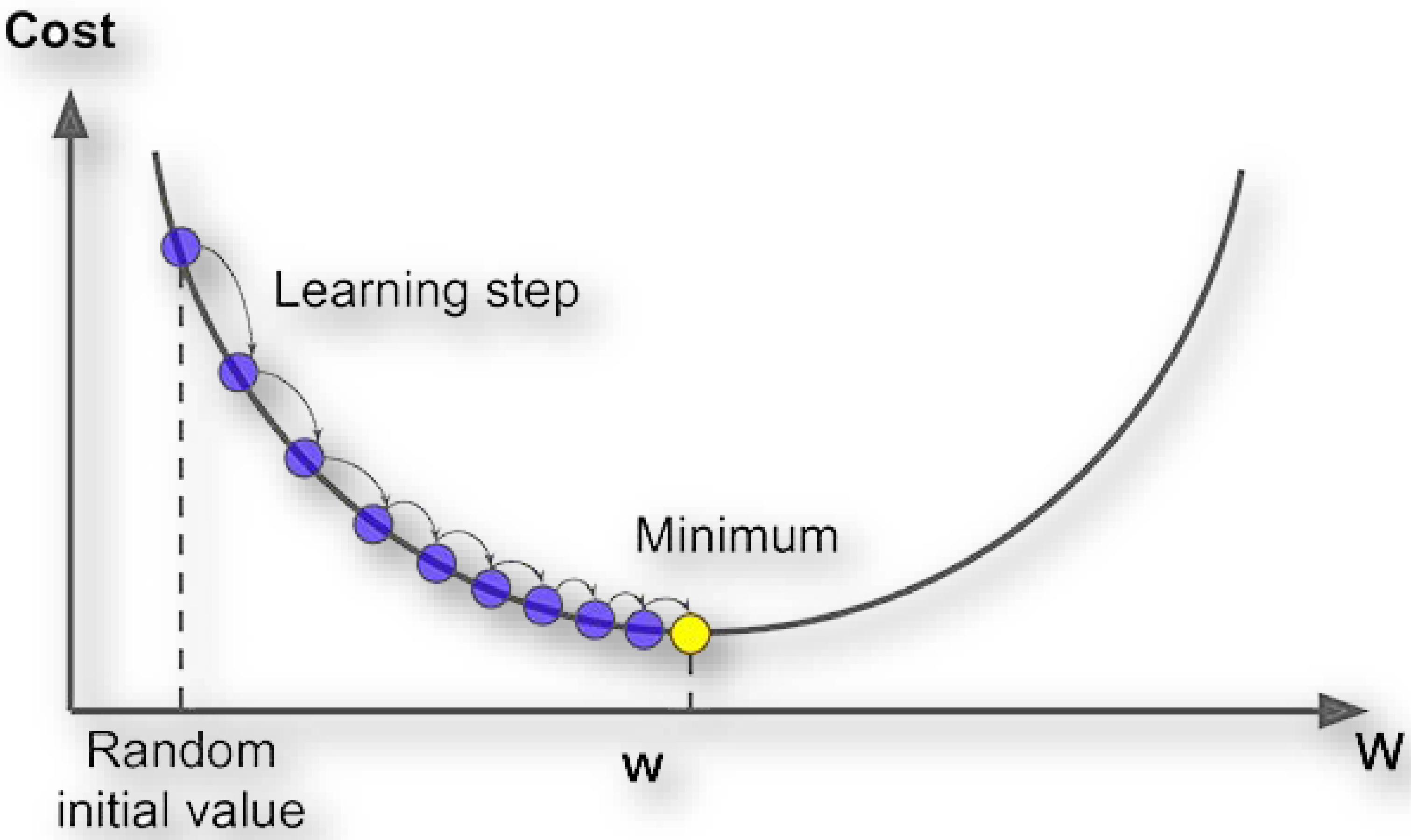
$$C = \frac{-1}{n} * \sum_{i=0}^{n} (y \ * \ln(a) + (1-y) \ * \ln(1-a))$$

y = actual values

a = predicted values

Gradient Descent

❖ It is an optimization algorithm for finding the minimum value of a function to reduce cost.



Back-Propagation

- It is simply a revision of weights.
- In this using Feed Forward Network, weights are changed till we get the lease cost function (error)